Diagonalization and the Minimal Polynomial

Problem sheet for the Day 7 class · CSIR NET previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET June 2016 MCQ Given a $n \times n$ matrix B define e^B by $e^B = \sum_{j=0}^{\infty} \frac{B^j}{j!}$. Let p be the characteristic polynomial of B. Then the matrix $e^{p(B)}$ is

(A) $I_{n\times n}$ (B) $0_{n\times n}$
(C) $eI_{n\times n}$ (D) $\pi I_{n\times n}$

2. CSIR NET Dec 2015 MCQ If A is a 5×5 real matrix with trace 15 and if 2 and 3 are eigenvalues of A, each with algebraic multiplicity 2, then the determinant of A is equal to

(A) 0 (B) 24
(C) 120 (D) 180

3. CSIR NET Dec 2018 MCQ Let A be an $n \times n$ complex matrix. Assume that A is self-adjoint and let B denote the inverse of $A + iI_n$. Then all eigenvalues of $(A - iI_n)B$ are

(A) purely imaginary (B) of modulus one
(C) real (D) of modulus less than one

4. CSIR NET June 2017 MSQ Which of the following 3×3 matrices are diagonalizable over $\mathbb{R}$?

(A) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$

(B) $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 4 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

5. CSIR NET June 2017 MSQ For any $n \times n$ matrix B, let $N(B) = \{X \in \mathbb{R}^n : BX = 0\}$ be the null space of B. Let A be a 4×4 matrix with $\dim(N(A - 2I)) = 2$, $\dim(N(A - 4I)) = 1$ and $\text{rank}(A) = 3$. Then

(A) 0, 2 and 4 are eigenvalues of A

(B) determinant$(A) = 0$

(C) A is not diagonalizable

(D) trace$(A) = 8$

6. CSIR NET June 2018 MSQ Let V be a vector space over $\mathbb{C}$ with dimension n. Let $T : V \to V$ be a linear transformation with only 1 as eigenvalue. Then which of the following must be true?

(A) $T - I = 0$

(B) $(T - I)^{n-1} = 0$

(C) $(T - I)^n = 0$

(D) $(T - I)^{2n} = 0$

7. CSIR NET June 2018 MSQ Let $A \in M_3(\mathbb{R})$ be such that $A^8 = I_{3\times 3}$. Then

(A) minimal polynomial of A can only be of degree 2

(B) minimal polynomial of A can only be of degree 3

(C) either $A = I_{3\times 3}$ or $A = -I_{3\times 3}$

(D) there are uncountably many A satisfying the above

8. CSIR NET Dec 2018 MSQ Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map that satisfies $T^2 = T - I_n$. Then which of the following are true?

(A) T is invertible

(B) $T - I_n$ is not invertible

(C) T has a real eigen value

(D) $T^3 = -I_n$

9. CSIR NET Dec 2018 MSQ Let A be a real matrix with characteristic polynomial $(X - 1)^3$. Pick the correct statements from below:

(A) A is necessarily diagonalizable

(B) If the minimal polynomial of A is $(X - 1)^3$, then A is diagonalizable

(C) Characteristic polynomial of A^2 is $(X - 1)^3$

(D) If A has exactly two Jordan blocks, then $(A - I)^2$ is diagonalizable

10. CSIR NET Dec 2016 MSQ Let n be an odd number ≥ 7. Let $A = [a_{ij}]$ be an $n \times n$ matrix with $a_{i,i+1} = 1$ for all $i = 1, 2, \ldots, n - 1$ and $a_{n,1} = 1$. Let $a_{ij} = 0$ for all the other pairs (i, j). Then we can conclude that

(A) A has 1 as an eigenvalue.

(B) A has -1 as an eigenvalue.

(C) A has at least one eigenvalue with multiplicity ≥ 2.

(D) A has no real eigenvalues.